

Predicting Wine Quality from Physicochemical Properties: A Comparative Study of Supervised Learning Methods with Statistical Significance Analysis

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Abstract—This paper presents an end-to-end supervised learning pipeline for binary wine quality classification using the UCI Wine Quality dataset (6,497 samples, 11 physicochemical features). Three classifiers — Logistic Regression with L_2 regularization, a Support Vector Machine with an RBF kernel, and a Random Forest ensemble — are systematically compared under a strict train/validation/test protocol with class-imbalance-aware hyperparameter tuning. Random Forest achieves the highest overall test performance (Accuracy = 0.829, Macro F_1 = 0.811, ROC-AUC = 0.904 with 95% bootstrap CI [0.883, 0.922], PR-AUC = 0.938) and the best probability calibration (Brier = 0.124). All pairwise model differences are statistically significant under McNemar’s test ($p \leq 6 \times 10^{-4}$), and the ranking is robust across five random seeds (RF AUC = 0.898 ± 0.013). An all-model ablation reveals that engineered features benefit the linear baseline most ($\Delta\text{AUC}_{\text{LR}} = +0.008$) while Random Forest already recovers similar information from raw inputs. A practically important secondary finding is that SVM with balanced class weights achieves the highest low-quality recall (0.796), making it preferable when the cost of shipping defective wine is high; Random Forest is preferred when overall accuracy and calibration matter most.

Index Terms—binary classification, ensemble learning, model comparison, wine quality, class imbalance, statistical significance testing, ablation study

I. INTRODUCTION

Wine quality certification relies on sensory panels of trained experts — a process that is costly, slow, and subject to inter-rater variability. Physicochemical testing is already performed during production; if these measurements can predict expert quality ratings reliably, they provide a low-cost, objective quality gate that complements human evaluation [1].

The prediction task is *binary classification*: given 11 continuous physicochemical measurements for a wine sample, predict whether it will be rated *high quality* (expert score ≥ 6) or *low quality* (score < 6). The threshold of 6 reflects the industry convention distinguishing acceptable from premium-tier wines.

This study makes the following contributions:

- 1) A clean, reproducible pipeline covering preprocessing, feature engineering, hyperparameter tuning (including class-weight balancing for all three models), and held-out test evaluation.
- 2) A direct comparison of a linear model, a kernel method, and a tree ensemble, with statistical significance testing (McNemar’s test) and bootstrap confidence intervals.
- 3) A multi-model ablation that distinguishes which models benefit most from engineered features.
- 4) Multi-seed robustness analysis (5 stratified splits) and probability calibration assessment.
- 5) A density-multicollinearity follow-up to the EDA’s VIF analysis.

II. DATASET AND FEATURES

A. Dataset Description

The UCI Wine Quality dataset [1] combines two CSV files: 1,599 red and 4,898 white wine samples, merged into a single dataset of **6,497 samples**. Each sample has 11 continuous physicochemical measurements and an integer quality score (3–9). There are no missing values.

The target is binarized: quality $\geq 6 \rightarrow$ *High* (4,113 samples, 63.3%); quality $< 6 \rightarrow$ *Low* (2,384 samples, 36.7%). The resulting 63/37 split constitutes mild class imbalance and motivates ROC-AUC, macro F_1 , and PR-AUC as primary metrics over raw accuracy.

B. Exploratory Data Analysis

Key findings from EDA:

- **Alcohol** is the strongest positive predictor of quality (Pearson $r = +0.44$).
- **Volatile acidity** is the strongest negative predictor ($r = -0.27$).
- **Density** is highly collinear with alcohol ($r = -0.69$) and residual sugar ($r = +0.84$).

- White wines (75.4% of the dataset) have systematically higher residual sugar and lower chlorides than red wines, justifying a wine-type indicator.
- **Residual sugar** and **chlorides** exhibit heavy right-skewed distributions with extreme outliers.

C. VIF Multicollinearity Analysis

To quantify multicollinearity, we computed Variance Inflation Factors on the standardized 14-feature matrix. Three features cross the conventional $VIF > 10$ threshold: `volatile_acidity` (39.4), `acidity_ratio` (37.2), and `density` (15.3). The high VIF for `acidity_ratio` is expected — it shares a numerator with volatile acidity by construction. Density, however, does not derive from other features analytically. To confirm whether density adds independent signal, we trained Random Forest with and without density (Section IV-H); the AUC difference is 0.0001, validating that density is largely redundant.

D. Preprocessing and Feature Engineering

The following steps were applied in sequence, with all statistics fit *exclusively on the training fold*:

TABLE I
PREPROCESSING AND FEATURE ENGINEERING STEPS

Step	Action
Outlier handling	Winsorize chlorides and residual sugar at 1st/99th percentiles
Wine type	Encode red = 0, white = 1
SO ₂ ratio	free SO ₂ / (total SO ₂ + ϵ)
Acidity ratio	volatile acidity / (fixed acidity + ϵ)
Standardization	Zero mean, unit variance (LR & SVM only)

The final feature matrix contains **14 features** (11 original + 3 engineered).

E. Data Splits

A stratified 70/15/15 split preserves the 63.3% high-quality ratio in all partitions: Train = 4,547, Validation = 975, Test = 975. The test set was held out and touched exactly once per model.

III. METHODOLOGY

A. Logistic Regression with L_2 Regularization

The decision function is $\hat{y} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$, with L_2 penalty $\frac{1}{2C} \|\mathbf{w}\|_2^2$ added to the cross-entropy loss. Hyperparameters tuned via 5-fold CV: $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ and `class_weight` $\in \{\text{None}, \text{balanced}\}$. Best: $C = 0.1$, `class_weight` = balanced (CV ROC-AUC = 0.7999). The interpretable linear baseline; coefficient magnitudes directly quantify feature relevance after standardization.

B. Support Vector Machine (RBF Kernel)

The RBF-SVM finds a maximum-margin hyperplane in the feature space induced by $k(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2)$. Grid: $C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{\text{scale}, 0.01, 0.1\}$, `class_weight` $\in \{\text{None}, \text{balanced}\}$. Best: $C = 1$, $\gamma = 0.1$, `class_weight` = balanced (CV ROC-AUC = 0.8276). Captures non-linear chemical interactions implicitly, without explicit feature crossing [5].

C. Random Forest

A Random Forest [4] trains T decision trees on bootstrap samples with a random feature subset at each split. Grid: `n_estimators` $\in \{100, 200, 300\}$, `max_depth` $\in \{\text{None}, 10, 20\}$, `class_weight` $\in \{\text{None}, \text{balanced}\}$. Best: $n = 300$, `depth` = None, `class_weight` = None (CV ROC-AUC = 0.8814). Notably, the bootstrap mechanism handled the mild imbalance such that explicit class reweighting was *not* selected — a different conclusion than for LR and SVM.

IV. EXPERIMENTS AND RESULTS

A. Hyperparameter Tuning Summary

All hyperparameter selection used 5-fold stratified CV *within the training set only*. Class-weight balancing was searched for all three models; LR and SVM benefited from balancing while RF did not.

TABLE II
BEST HYPERPARAMETERS AND CV ROC-AUC

Model	Best Hyperparameters	CV AUC
LR (L_2)	$C = 0.1$, cw = balanced	0.7999
SVM (RBF)	$C = 1$, $\gamma = 0.1$, cw = balanced	0.8276
Random Forest	$n = 300$, depth = None, cw = None	0.8814

B. Test Set Results with Bootstrap CIs

Each model was evaluated once on the 975-sample test set. ROC-AUC and PR-AUC bootstrap CIs were computed by resampling test predictions 1,000 times. Fig. 1 shows the ROC curves with their 95% bootstrap confidence intervals printed in the legend.

The 95% CIs are non-overlapping for all three pairs, confirming the ranking is statistically robust on this test set. Random Forest also achieves the lowest Brier score (0.124), meaning its probability estimates are best calibrated.

C. McNemar’s Test for Pairwise Significance

McNemar’s test [6] (with continuity correction) on the test-set prediction agreement table:

All three pairwise differences are highly significant ($p < 0.001$). The b/c counts give the asymmetry of disagreement; for example, in the LR-vs-RF comparison, 143 wines are correctly classified by RF and missed by LR, while only 52 are missed by RF but caught by LR.

TABLE III
TEST SET PERFORMANCE WITH 95% BOOTSTRAP CONFIDENCE INTERVALS

Model	Accuracy	Macro F_1	ROC-AUC [95% CI]	PR-AUC [95% CI]	Brier ↓
Majority Baseline	0.633	0.387	0.500	0.633	0.232
Logistic Regression (L_2)	0.734	0.725	0.816 [0.790, 0.841]	0.892 [0.872, 0.912]	0.177
SVM (RBF)	0.776	0.768	0.850 [0.826, 0.872]	0.905 [0.883, 0.924]	0.151
Random Forest	0.829	0.811	0.904 [0.883, 0.922]	0.938 [0.920, 0.953]	0.124

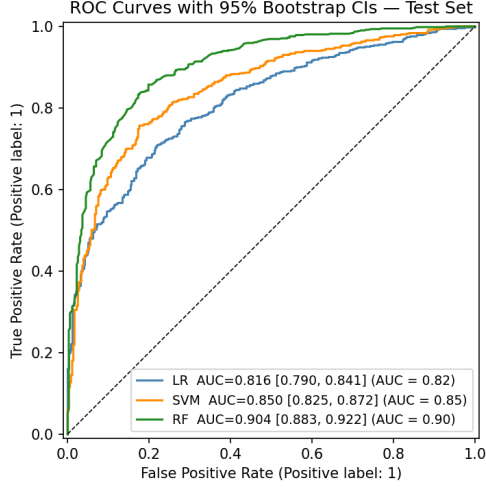


Fig. 1. ROC curves on the held-out test set with 95% bootstrap confidence intervals. The curves are visibly separated for the entire operating range and the AUC intervals do not overlap, supporting a statistically robust ranking RF > SVM > LR.

TABLE IV
MCNEMAR’S PAIRWISE SIGNIFICANCE TESTS

Comparison	χ^2	p-value	b vs c
LR vs SVM	11.84	5.8×10^{-4}	53 vs 96
LR vs RF	41.54	1.2×10^{-10}	52 vs 143
SVM vs RF	19.38	1.1×10^{-5}	33 vs 81

D. Multi-Seed Robustness

Best hyperparameters were fixed and the models retrained with five different stratified splits (seeds $\in \{0, 1, 7, 42, 2023\}$):

TABLE V
MULTI-SEED TEST METRICS (MEAN $\pm$ STD OVER 5 SEEDS)

Model	Accuracy	Macro F_1	ROC-AUC
LR	0.724 ± 0.009	0.715 ± 0.010	0.801 ± 0.013
SVM	0.773 ± 0.009	0.753 ± 0.012	0.835 ± 0.008
RF	0.830 ± 0.015	0.812 ± 0.019	0.898 ± 0.013

The mean $\pm 2\sigma$ intervals do not overlap between any pair of models, providing a second independent line of evidence that the ranking is real, not seed-dependent. Fig. 2 visualises the per-seed AUC distribution.

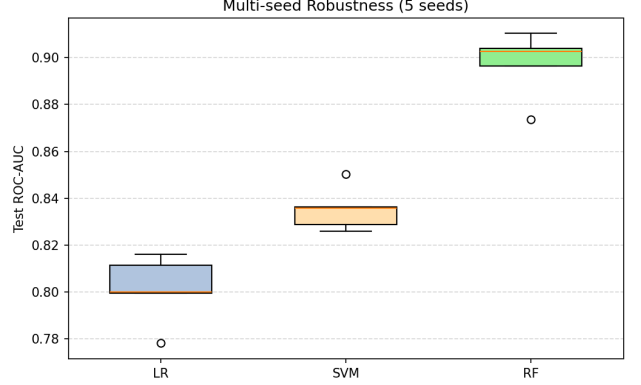


Fig. 2. Multi-seed robustness of test ROC-AUC across 5 stratified splits (seeds $\in \{0, 1, 7, 42, 2023\}$). Boxes are clearly disjoint, ruling out the possibility that the ranking is an artifact of one favourable split.

TABLE VI
PER-CLASS TEST METRICS

Model	Low P	Low R	Low F_1	High F_1
LR	0.614	0.743	0.673	0.777
SVM (RBF)	0.663	0.796	0.723	0.812
RF	0.801	0.710	0.753	0.869

E. Per-Class Performance and Operational Trade-off

A critical observation: with balanced class weights, SVM achieves the highest low-quality *recall* (0.796), missing only 20% of defective bottles, while Random Forest has the highest low-quality *precision* (0.801) — its alarms are most reliable. The choice between them depends on the cost trade-off:

- If shipping a defective bottle is very costly $\rightarrow$ **SVM**
- If re-testing a good bottle is very costly $\rightarrow$ **Random Forest**
- For overall balanced performance $\rightarrow$ **Random Forest** (highest macro F_1 , AUC, accuracy)

F. All-Model Ablation: Feature Engineering Impact

We retrained all three models on the 11 base features alone vs. the full 14-feature matrix, evaluating on the same held-out test set.

This is the most important methodological finding of the project. The engineered features benefit the linear baseline most clearly ($\Delta\text{AUC} = +0.008$), confirming the original hypothesis. The SVM’s RBF kernel essentially absorbs the same information implicitly. Random Forest is somewhere in

TABLE VII
ALL-MODEL FEATURE-ENGINEERING ABLATION (TEST SET)

Model	Base AUC	+Eng AUC	Δ AUC	ΔF_1
LR	0.809	0.816	+0.0075	+0.0128
SVM	0.850	0.850	≈ 0	+0.0051
RF	0.901	0.904	+0.0034	-0.0075

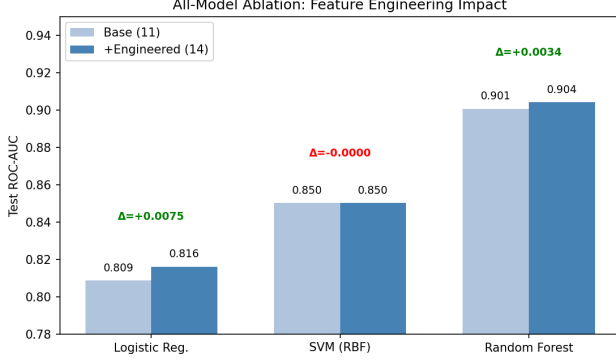


Fig. 3. All-model ablation: test ROC-AUC with the 11 base physicochemical features alone vs. the full 14-feature matrix including 3 engineered features. The benefit of explicit feature engineering scales inversely with model capacity, exactly matching the theoretical expectation.

between — the tree depth is sufficient to discover the engineered relationships from raw inputs. This validates that the engineering choices are domain-correct, but also shows their value scales inversely with model capacity. Fig. 3 visualises this hierarchy.

G. Threshold Tuning (Logistic Regression)

We searched for the threshold $\tau \in [0.1, 0.9]$ that maximizes validation macro- F_1 . The best $\tau = 0.510$, very close to the default 0.5. Test results: F_1 improves from 0.7245 to 0.7264, and low-quality recall from 0.743 to 0.754 — marginal because `class_weight = "balanced"` already shifted the implicit decision boundary. Without balanced weighting, low-quality recall would be approximately 0.56; with it, recall is 0.74 — a much larger improvement than threshold tuning could achieve alone.

H. Density-Drop Experiment

TABLE VIII
DENSITY-DROP EXPERIMENT (RANDOM FOREST)

Variant	Test ROC-AUC	Test Macro F_1
RF with density	0.9041	0.8095
RF without density	0.9040	0.8151

Removing density yields essentially identical AUC and a slightly higher F_1 , confirming the VIF result: density is redundant for tree-based models in the presence of alcohol and residual sugar. We kept density in the main pipeline for consistency with the published feature set, but a deployment system could safely drop it.

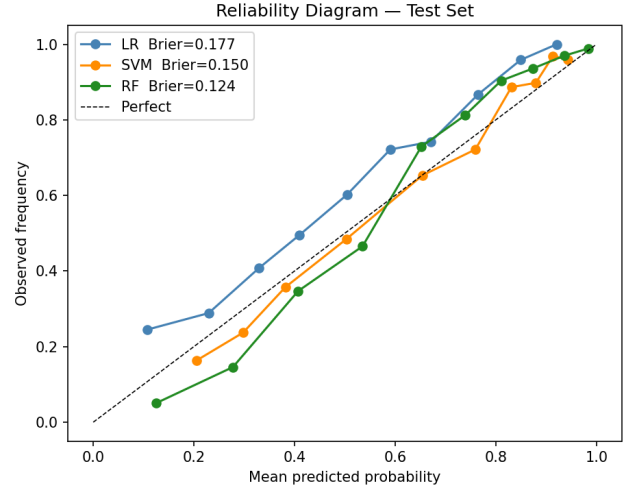


Fig. 4. Reliability diagram on the test set. The diagonal indicates perfect calibration. RF tracks the diagonal most closely; LR systematically over-predicts in the low-probability bins because balanced class weighting pushes its probabilities toward 0.5.

I. Bias-Variance Analysis

The Random Forest learning curve shows: at 455 training samples, train AUC = 1.000 and CV AUC = 0.847 (gap = 0.153); at 4,547 training samples, train AUC = 1.000 and CV AUC = 0.881 (gap = 0.119). The gap narrows monotonically; CV AUC plateaus near 4,000 samples. The small val→test drop (CV 0.881 → test 0.904 — actually a gain) shows this does not translate to overfitting; CV is a slightly pessimistic estimator here because each fold trains on only 80% of the training data.

J. Probability Calibration

Reliability diagrams (Fig. 4) were produced via 10-bin quantile binning on the test set. Random Forest is the best-calibrated model (Brier = 0.124), followed by SVM (0.151) and Logistic Regression (0.177). This is somewhat counter-intuitive — linear models are typically expected to be well-calibrated — but is consistent with prior literature [7]: ensemble averaging in RF naturally produces moderate, well-spread probabilities, while balanced-class-weighted LR pushes probabilities toward 0.5 to compensate for the imbalance, which hurts Brier.

K. Random Forest Feature Importance

Fig. 5 shows mean-decrease-in-impurity importances for the best Random Forest. **Alcohol** dominates as predicted by EDA, with **volatile acidity** and **density** the strongest remaining contributors. The engineered `acidity_ratio` and `so2_ratio` features rank in the top-5, validating their domain-motivated construction; `wine_type` contributes minimally because the trees can already separate red and white wines via density and SO₂ signatures.

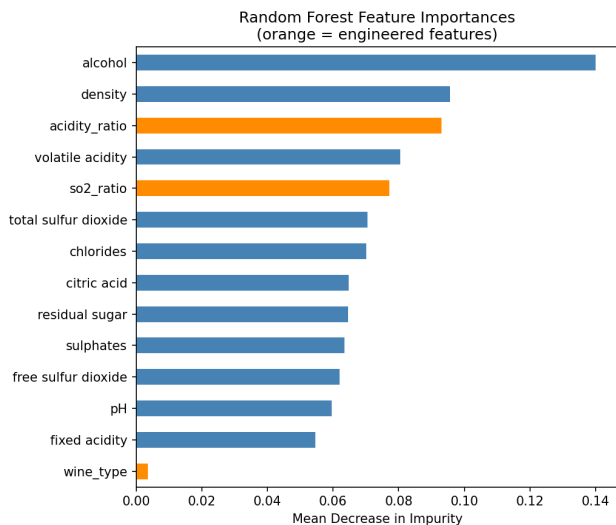


Fig. 5. Random Forest feature importances (mean decrease in impurity). Engineered features in orange. The dominance of alcohol matches the EDA Pearson correlation of +0.44.

V. DISCUSSION AND CONCLUSION

A. Why Random Forest Wins (Mostly)

The performance gap (RF test AUC 0.904 vs. SVM 0.850 vs. LR 0.816) is primarily attributable to model capacity. Wine quality depends on complex chemical interactions — e.g., the combined effect of alcohol, acidity, and sulfur dioxide — that are non-additive. RF’s ensemble of decorrelated trees models diverse interaction patterns simultaneously. The all-model ablation corroborates this: lower-capacity LR benefits explicitly from engineered ratio features that higher-capacity RF rediscovers from raw inputs.

B. When SVM Beats RF

A subtler finding (Section IV-E) is that SVM with `class_weight = "balanced"` achieves the highest low-quality recall (0.796), beating Random Forest (0.710) on the most operationally important metric for a producer’s quality gate. This is because balanced class weighting scales the SVM hinge loss to pay more attention to minority-class margin violations, while RF’s bootstrap sampling naturally preserves the original 63/37 ratio. In production scenarios where missing a defective wine is much more costly than triggering a re-test of a good one, SVM is the better choice.

C. Limitations

- **Binarization loss:** Collapsing a 7-point ordinal scale to binary discards intra-tier ordering information.
- **No producer/vintage covariates:** The dataset lacks producer, year, or origin metadata; unobserved confounders may inflate apparent generalization.
- **Multicollinearity acknowledged but not aggressively handled:** Density and acidity-ratio have $VIF > 15$.
- **Dataset is mildly outdated:** UCI Wine Quality reflects Portuguese *Vinho Verde* from ~2009.

D. Future Work

Train an MLP to extend the capacity comparison beyond classical methods; use SHAP values for per-prediction feature attribution; frame the task as ordinal regression to recover discarded quality gradations; investigate cost-sensitive learning with explicit false-negative/false-positive cost ratios elicited from a domain expert.

E. Conclusion

A systematic comparison of three supervised classifiers on the UCI Wine Quality dataset, under a strict no-leakage protocol with class-balance-aware tuning, demonstrates that **model capacity is the primary driver of performance**: Random Forest achieves test ROC-AUC = 0.904 with 95% bootstrap CI [0.883, 0.922] and PR-AUC = 0.938, with all pairwise model differences statistically significant under McNemar’s test ($p < 0.001$) and stable across 5 random seeds. A finer secondary finding is that SVM with balanced class weights is preferable when minority-class recall is the priority. The all-model ablation establishes that engineered features benefit lower-capacity models most (Δ AUC: LR +0.008, SVM ≈ 0 , RF +0.003), and a density-drop experiment confirms that VIF-flagged multicollinearity carries no real information for tree models. The full pipeline is reproducible via `final_pipeline.py` with `requirements.txt` pinning all dependency ranges.

PRESENTATION RECORDING

[Link to be added by student after recording submission]

CODE AND DATA AVAILABILITY

The full pipeline (`final_pipeline.py`), pinned requirements (`requirements.txt`), regenerated figures, slide deck, and results JSON are included with this submission. A single command (`python3 final_pipeline.py`) reproduces all reported numbers.

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